Question

Ammonium sulfide is a widely used qualitative analytical reagent. To prepare this reagent, hydrogen sulfide gas is introduced into 4~5 mol/L aqueous ammonia, followed by the addition of water. Pure solutions cannot be obtained by this method: when absorbing gas below or above the stoichiometric amount, the solution will contain excess ammonia or ammonium hydrosulfide.

Dilute 10.00 mL of the ammonium sulfide test solution to 1.000 L, then transfer 10.00 mL of the resulting stock solution to a distillation flask and add approximately 40 mL of water. Next, add 25.00 mL of 0.1 mol/L cadmium nitrate solution to the collection flask (where the distilled components will condense), and introduce 20.00 mL of 0.02498 mol/L sulfuric acid into the distillation flask.

Distill approximately half of the solution from the distillation flask into the collection flask. (A yellow precipitate can be observed in the collection flask.)

Thoroughly wash the remaining contents in the distillation flask into a conical flask, add a few drops of methyl red indicator, and titrate with 0.05002 mol/L NaOH solution, consuming 10.97 mL of NaOH at the endpoint.

Add bromine water to the solution in the collection flask (the precipitate dissolves), heat and boil the solution for 15 minutes to remove excess bromine. The bromine oxidizes all sulfur-containing ions to sulfate ions. Titrate the solution in the collection flask with 0.1012 mol/L NaOH, consuming 15.01 mL.

Calculate the pH of this reagent solution and select the closest option.

$$\mathrm{NH_3} : \mathrm{K_b} = 1.76\times10^{-5}; \mathrm{H_2S} : \mathrm{K_{a1}} = 1.1\times10^{-7}, \mathrm{K_{a2}} = 1.3\times10^{-13}$$

A. 3.72

B. 4.13

C. 4.89

D. 5.02

E. 6.27

F. 7.33

G. 7.92

H. 8.54

I. 9.54

J. 10.06

K. 10.54

L. 11.46

M. 12.09

N. 12.68

O. 13.38

Answer

Correct Answer: I

Detailed Explanation

Ions remaining in the distillation flask are NH_4^+ and H^+, and only H^+ is consumed when methyl red is used as an indicator.

CHECKPOINT 1 PTS

Only H^+ is titrated when methyl red is used as an indicator

The remaining NH_4^+ is produced by the protonation of NH_4^+ by H_2SO_4, therefore:

$$c(NH_3) = (0.02000 \times 0.02498 \times 2 - 0.05002 \times 0.01097) \times \frac{100}{0.01000} = 4.505(M)$$

CHECKPOINT 1 PTS

Ammonium sulfide reagent solution contains 4.505 M NH_3

Two reactions occurred in the receiving flask:

$$Cd^2 + +H_2S = CdS + 2H^+$$

$$CdS + 4Br_2 + 4H_2O = CdSO_4 + 8HBr$$

Therefore, one equivalent of H_2S produces ten equivalents of H^+.

CHECKPOINT 1 PTS

$H_2S - 10H^+$

Thus: $c(H_2S) = 0.1012 \times 0.01501 \times 0.1 \times \frac{100}{0.01000} = 1.519(M)$

That is, $c(\text{totalS}) = 1.519\text{M}$ in the reagent.

CHECKPOINT 1 PTS

$c(\text{totalS}) = 1.519\text{M}$

$c(NH_4HS) = 4.505 - 1.519 = 2.986\text{M}$

CHECKPOINT 1 PTS

$c(NH_3) = 4.505 - 1.519 = 2.986\text{M}$

Charge balance:

$[NH_4^+] + [H^+] = 2[S^{2-}] + [HS^-] + [OH^-]$

CHECKPOINT 1 PTS

$[NH_4^+] + [H^+] = 2[S^{2-}] + [HS^-] + [OH^-]$

Substituting into the distribution fraction expression, we get $[\mathrm{H}^+] = 2.88\times10^{-10}\mathrm{M}$

CHECKPOINT 1 PTS

$[\mathrm{H}^+] = 2.88\times10^{-10}\mathrm{M}$

pH = 9.54

CHECKPOINT 1 PTS

pH = 9.54